

GANTT CHART

Introduction

A Gantt Chart is a project management tool used to plan, schedule, and monitor the progress of a project over a specific period. It was developed by Henry L. Gantt in the early 1900s and is one of the most widely used techniques for project scheduling.

A Gantt chart represents project activities on a timeline using horizontal bars. Each bar corresponds to a specific task and indicates its start date, end date, duration, and progress. It helps project managers and team members understand what tasks need to be completed, when they should be completed, and how different activities are related to one another. The chart also makes it easier to identify delays, monitor project status, allocate resources efficiently, and ensure that the project is completed within the planned schedule.

In Software Engineering, Gantt charts are commonly used during project planning to divide the software development life cycle into manageable phases such as requirement gathering, design, development, testing, and deployment. By tracking each activity and its dependencies, the development team can maintain better coordination and complete the project on time.

Components of a Gantt Chart

A Gantt chart consists of several important components that help in planning and tracking the progress of a project.

1. Task/Activity

A task is an individual piece of work that must be completed during the project. Each task is listed in chronological order on the left side of the Gantt chart.

Example: Gather Requirements, System Design, Development, Testing.

2. Timeline

The timeline is displayed at the top of the chart and represents the project's duration in terms of days, weeks, or months. It helps determine when each task starts and ends.

3. Start Date

The start date indicates the planned date on which a particular task begins.

4. End Date

The end date specifies the planned completion date of a task.

5. Duration

Duration represents the total time required to complete a task. It may be measured in hours, days, weeks, or months depending on the project.

6. Progress (% Complete)

This component shows how much of a task has been completed. It helps project managers monitor the project's current status.

Example: 25%, 50%, 75%, or 100%.

7. Milestone

A milestone is a significant event or checkpoint that marks the completion of a major phase of the project. It has no duration and is usually represented by a diamond symbol (◆) in a Gantt chart.

Example: Requirements Approved, Design Completed, Project Completed.

8. Dependencies

Dependencies indicate the relationship between tasks. Some tasks cannot begin until previous tasks are completed. This helps maintain the correct sequence of activities.

Example: Development starts only after the design phase is completed.

9. Task Bar

A task bar is the horizontal bar used to represent the duration of each task on the timeline. The length of the bar indicates how long the task will take.

10. Resource Assignment (Optional)

This component identifies the person, team, or department responsible for completing a particular task. It helps in resource planning and workload management.

11. Status Indicator

Some Gantt charts include a status indicator (such as On Schedule, Delayed, or Completed) to quickly show the current state of each activity.

12. Critical Path (Optional)

The critical path is the sequence of tasks that directly affects the project's completion date. Any delay in these tasks will delay the entire project.

Advantages of a Gantt Chart

- Provides a clear visual representation of the entire project schedule.
- Helps in effective planning and scheduling of tasks.
- Makes it easy to track project progress.

- Clearly identifies task dependencies and sequence.
- Improves communication among project stakeholders.
- Assists in efficient resource allocation.
- Helps identify delays and take corrective actions early.
- Ensures timely completion of project activities.

Disadvantages of a Gantt Chart

- Becomes difficult to manage for very large or complex projects.
- Requires frequent updates when project schedules change.
- Does not represent task complexity or effort in detail.
- Can become cluttered if too many tasks are included.
- Does not automatically resolve resource conflicts.
- Creating and maintaining detailed Gantt charts can be time-consuming.

Planning of AI Recruitment and Job Management System Using Gantt Chart

The Gantt chart for the AI Recruitment and Job Management System outlines the complete development timeline of the project, divided into five major phases: **Requirements and Planning, System Design, Development, Testing, and Deployment.**

Each phase is further divided into specific tasks with defined timelines and milestones, allowing the development team to systematically track the project from initial planning to final deployment.

The activities cover requirements gathering, system and database design, implementation of recruitment features, AI-based resume parsing and candidate matching, application tracking, testing, and final deployment.

1 Requirements and Planning

This is the foundational phase of the AI Recruitment and Job Management System, where the project scope and requirements are identified and finalized.

The activities include the project kickoff and scope definition, requirements gathering and preparation of the Software Requirements Specification (SRS), and review and approval of the requirements.

This phase establishes the functional and non-functional requirements of the system and provides a clear foundation for the design and development activities that follow.

The SRS covers important functions such as candidate management, job posting, AI-based resume parsing and matching, application tracking, notifications, and recruiter analytics.

2 System Design

In this phase, the technical structure and design of the AI Recruitment and Job Management System are prepared.

The activities include creating the Use Case Diagram, Data Flow Diagram (DFD), and Activity Diagram, followed by the development of the Entity Relationship Diagram (ERD) and Class Diagram.

The team then prepares the database schema design and develops the UI wireframes, followed by a final design review and design freeze.

These activities provide the blueprint for the system architecture, database structure, user interactions, and interface before implementation begins.

The system is planned as a three-tier architecture consisting of the user-facing application, backend and AI/NLP services, and persistent data storage.

3 Development

The development phase is the main implementation stage of the project, where the core features of the AI Recruitment and Job Management System are developed.

The activities include implementing candidate registration and login, the recruiter dashboard and job posting module, resume upload and AI-based resume parsing, the Application Tracking System (ATS), notifications, and analytics.

The AI module parses resumes into structured information and calculates candidate-job match scores using semantic similarity.

The ATS manages application statuses such as **Applied, Shortlisted, Interview Scheduled, Rejected, and Hired**.

This phase concludes with the completion of the major system development activities and preparation for testing.

4 Testing

Once the development activities are completed, the testing phase is carried out to ensure that the system works correctly and satisfies the defined requirements.

This phase includes unit and integration testing, system and website testing, and user acceptance testing (UAT) with a prototype demonstration.

Testing focuses on verifying the individual modules as well as their integration, including candidate registration, job posting, resume parsing, AI matching, application tracking, notifications, and analytics.

Any problems identified during testing are documented and addressed before the system proceeds to deployment.

The testing phase helps ensure that the system is reliable, usable, secure, and ready for final release.

5 Deployment

The final phase focuses on preparing the AI Recruitment and Job Management System for actual use.

It includes bug fixes and final refinements based on the results of testing and feedback, followed by the deployment and go-live of the system.

The deployment stage marks the completion of the development lifecycle and makes the system ready for use by candidates, recruiters, and administrators.

The project also considers cloud deployment and CI/CD as part of its planned technical environment.

GANTT CHART'S

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